

Products

Solutions

Services

Training Level Radar_Envelope curve & mapping



Slide 1

12/12/2013

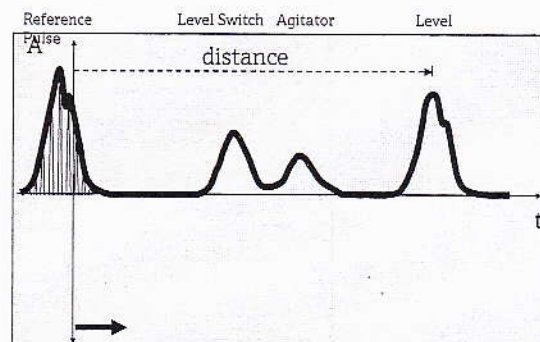
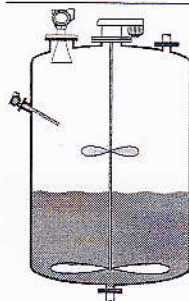
Muflich

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Applied Process Automation

Training Level Radar_Envelope curve & mapping

Creation of Envelope Curve



- Amplification and logarithmic calculus of the received signals
- Rectification of the emitted pulse and the received signals
- Generation of the Envelope Curve
- Distance evaluation

Slide 2

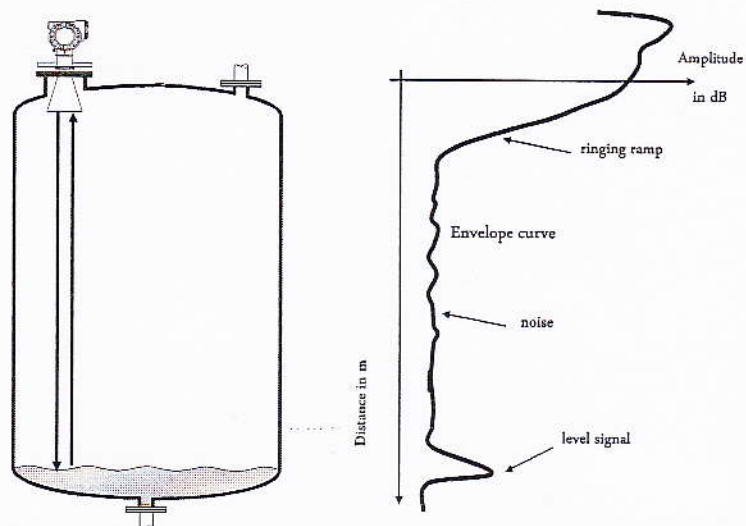
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Training Level Radar_Envelope curve & mapping

Process Tank <-> Envelope Curve



Slide 3

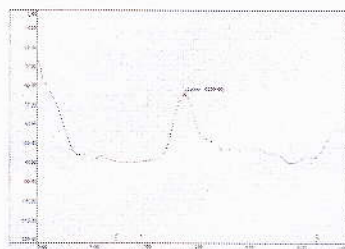
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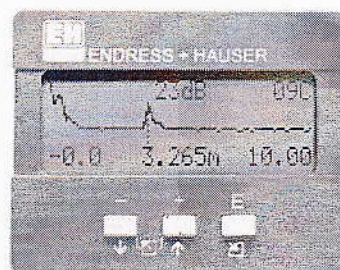
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Different Ways to Display the Envelope Curve



FieldCare™ Service Tool
(e.g. CompuBox FXA291)



Micropilot device display

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Additional Curves I - Overview

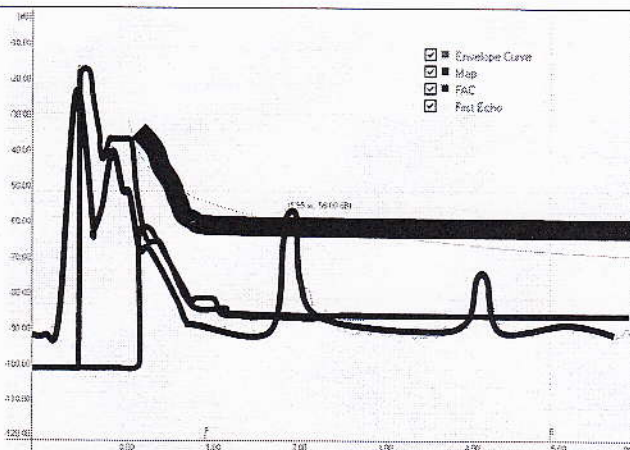


Figure 1: The Envelope Curve shows the signal intensity of the radar signal. The Map curve shows the signal intensity of the radar signal. The FAC curve shows the signal intensity of the radar signal. The Fast Echo curve shows the signal intensity of the radar signal.

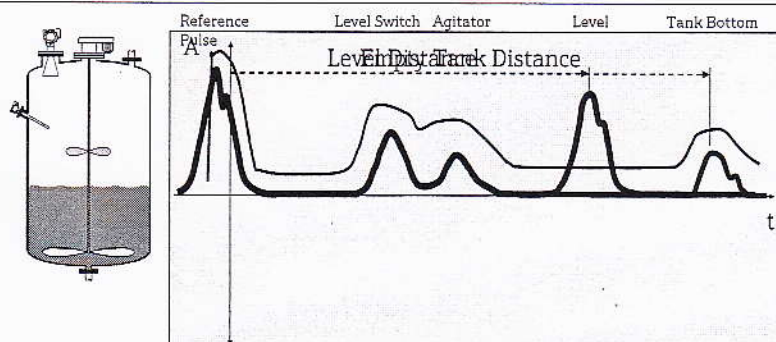
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Additional Curves II - Map I



- Map has to be recorded with a completely empty tank
- With the map, all static tank obstacles are getting faded out
- Only the actual level signal reaches over the map, all other echoes are not getting evaluated

Slide 6

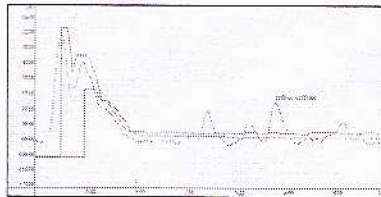
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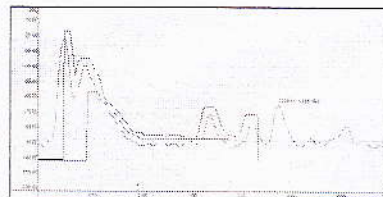
Additional Curves II – Map II

- Tasks of the Map:
 - tolerance threshold level signal detection
 - suppression of static interference signals



Factory map:

- suppression of false echo detection caused by reflections inside the RADAR device
- recorded in the factory
- has to be recorded again after hardware exchange



Customer map:

- suppression of false echo detection caused by reflections at static obstacles in a tank
- has to be recorded after initial device installation

Any questions?